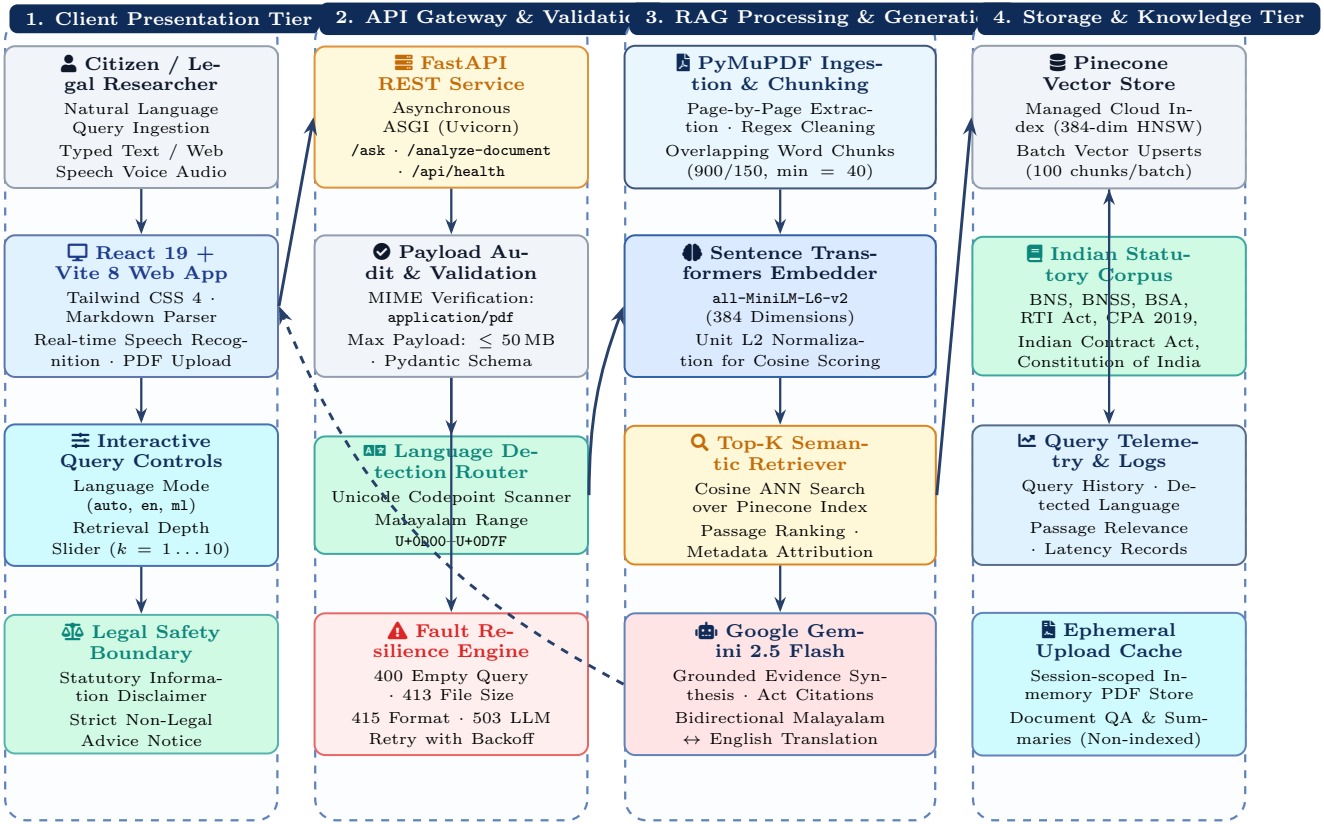

1. Client Presentation Tier
2. API Gateway & Validati
3. RAG Processing & Generati
4. Storage & Knowledge Tier
Citizen / Legal Researcher
Natural Language Query Ingestion
Typed Text / Web Speech Voice Audio
React 19 + Vite 8 Web App
Tailwind CSS 4 · Markdown Parser
Real-time Speech Recognition · PDF Upload
Interactive Query Controls
Language Mode (auto, en, ml)
Retrieval Depth Slider (k = 1 . . . 10)
Legal Safety Boundary
Statutory Information Disclaimer
Strict Non-Legal Advice Notice
FastAPI REST Service
Asynchronous ASGI (Uvicorn)
/ask · /analyze-document · /api/health
Payload Audit & Validation
MIME Verification: application/pdf
Max Payload: ≤ 50 MB · Pydantic Schema
Language Detection Router
Unicode Codepoint Scanner
Malayalam Range U+0D00–U+0D7F
Fault Resilience Engine
400 Empty Query · 413 File Size
415 Format · 503 LLM Retry with Backoff
PyMuPDF Ingestion & Chunking
Page-by-Page Extraction · Regex Cleaning
Overlapping Word Chunks (900/150, min = 40)
Sentence Transformers Embedder
all-MiniLM-L6-v2 (384 Dimensions)
Unit L2 Normalization for Cosine Scoring
Top-K Semantic Retriever
Cosine ANN Search over Pinecone Index
Passage Ranking · Metadata Attribution
Google Gemini 2.5 Flash
Grounded Evidence Synthesis · Act Citations
Bidirectional Malayalam ↔ English Translation
Pinecone Vector Store
Managed Cloud Index (384-dim HNSW)
Batch Vector Upserts (100 chunks/batch)
Indian Statutory Corpus
BNS, BNSS, BSA, RTI Act, CPA 2019,
Indian Contract Act, Constitution of India
Query Telemetry & Logs
Query History · Detected Language
Passage Relevance · Latency Records
Ephemeral Upload Cache
Session-scoped In-memory PDF Store
Document QA & Summaries (Non-indexed)